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EDUCATION

COMSATS University, Lahore | *Bachelor of Science in Software Engineering* 2021 – 2025

International School Lahore, DHA Campus | *A-Levels (Advanced Level)* 2019 – 2021

EXPERIENCE

Freelance Full-Stack Developer 2025 – Present
Self-Employed Lahore, Pakistan

- Built **responsive websites and applications** using React, Next.js, Astro, Node.js and Python.
- Developed **REST APIs**, authentication, database-backed features and third-party integrations with **PostgreSQL and MongoDB**.
- Handled deployment using **Cloudflare, AWS and GitHub Actions**, working directly with clients from requirements through review and delivery.

Punjab Safe Cities Authority (PSCA) [🔗](#) Jun 2024 – Aug 2024
Web Development Intern Lahore, Pakistan

- Supported maintenance of **citizen-facing government portals**, working on **Node.js** services behind high-traffic provincial systems.
- Traced the data pipeline of Punjab's **Automated E-Challan Network** end to end, from AI-based camera detection through to web-dashboard visualization.

PROJECTS

Kitty – On-Site AI Agent [🔗](#) | *Python, FastAPI, LangGraph, PostgreSQL / pgvector* [GitHub](#) | [Case Study](#)

- Built a tool-calling agent on a **hand-built LangGraph loop** with **SSE streaming**, bounded tools, clarification interrupts and **PostgreSQL**-backed conversations, tested against a **67-case evaluation set**.

Replywork – Support Inbox Backend [🔗](#) | *TypeScript, Fastify, PostgreSQL, Gemini* [GitHub](#)

- Built a backend that answers product questions in Crisp and Chatwoot from approved **PostgreSQL** records, with signed webhooks, deduplicated queueing, retries and **human takeover**, covered by **220 offline tests**.

CUI Central – Campus Ecosystem [🔗](#) | *Next.js, Socket.IO, Gemini AI, MongoDB* [Live Site](#)

- Architected a multi-tenant campus ERP consolidating **7+ core services** (Digital Cafeteria, Job Portal, Room Booking) into a single **Next.js 15** monorepo. Implemented **RBAC** (Role-Based Access Control) for Admin, Faculty and Student workflows.
- Engineered a real-time layer with **Socket.IO** for live faculty availability and order status updates. Integrated a **RAG-based AI agent** (Google Gemini) with MongoDB Vector Search to handle student support queries.

Cloud Visitor Counter – Serverless Backend [🔗](#) | *AWS, Terraform, Python, GitHub Actions* [GitHub](#)

- Built a serverless backend tracking live visitor counts on AWS (Lambda, API Gateway, DynamoDB), with all infrastructure provisioned in **Terraform** (IaC).
- Automated deployment through **GitHub Actions** with keyless **OIDC** authentication, with atomic counter increments and remote state management.

Check! – Multiplayer Game [🔗](#) | *Next.js, Node.js, Socket.IO, XState* [Live Site](#) | [GitHub](#)

- Designed a custom WebSocket game engine supporting concurrent rooms, holding **<50ms latency** for game state sync over event-driven **Socket.IO**. Used **XState** state machines to manage game logic and remove race conditions across multiple clients.

Karting Analysis – Lap-Time Pipeline [🔗](#) | *Python, Pandas, NumPy, Matplotlib* [Live Site](#) | [GitHub](#)

- Built a Python pipeline to ingest and clean **7,500+ leaderboard records** from two tracks, cutting processing time by **30%** by replacing loops with **NumPy** vectorization, then charted lap-time distributions.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Web & Frameworks: Next.js, React, Astro, Node.js, Express, Fastify, FastAPI, Socket.IO, Tailwind CSS

Data & AI: Pandas, NumPy, Matplotlib, LangGraph, RAG, Vector Search, LLM Evals, Gemini API

Cloud & Tools: AWS, Terraform, Cloudflare, Docker, Git, GitHub Actions, MongoDB, PostgreSQL, Linux